Berk Yilmaz

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Experience

Software Developer Dec 2024 – Present
Vretta (Systems team) *Toronto, ON*

- Led performance testing end to end for fixed-date online exams serving **195,000+ students** across **4 provinces**, from environment builds and **k6** scripting on **Kubernetes (AWS EKS)** to client reports.
- Built production-scale test environments from **AWS RDS** snapshots, anonymizing **177,130** student IDs and birth dates the old script would have missed, then seeding students, assessments and test windows.
- Raised load-test coverage of production traffic from **60% to 92%** on a provincial program by recording student flows and checking **Node.js** and **Feathers.js** API payloads against **ALB** logs in **Athena**, catching empty 200s.
- Scaled simulated exams to **8,000 concurrent students** across API, WebSocket, **Redis** and database layers, finding **16%** of exam starts failed at **5**× production's login rate.
- Lowered planned infrastructure cost for **2** provincial exam periods by an estimated **26–32%** by benchmarking **AWS RDS** and **EC2** configurations under load, halving the planned database in detailed capacity reports.
- Derived a capacity rule of one API server per **1,000 students** a day from **k6** and **CloudWatch** data that matched 2025 production within **1%**; production then ran **12,800 students** a day on **15** servers with room to spare.
- Engineered role-based access control across **35** API services and **52 Angular** components of a provincial exam portal, adding **11** role-specific **Feathers.js** endpoints and **32** hooks to separate school, district and ministry data.
- Patched **5** authentication vulnerabilities in a student-data **Node.js** API, shipping fixes to production for **2** clients.
- Traced **2,781** production errors in an exam's worst hour to a single application defect using **CloudWatch**.
- Built out the production error-alerting pipeline to **2** more provinces on **AWS Lambda**, **Athena**, **Firehose** and **DynamoDB** with **GitLab CI**, keeping each province's data in its own AWS region.

Full-Stack Web Developer May 2024 – Dec 2024
Expense Trend (Contract) *Calgary, AB*

- Integrated **Plaid**, **MX** and **Mastercard** bank APIs in **Node.js** after their vendor reviews, connecting users of a **5-star** iOS app to **100+** Canadian financial institutions and automating personal and business expense entry.
- Built **6 OpenAPI**-documented endpoints using **Node.js** and **MSSQL** for community profiles and bank status, supporting tailored deals for newcomers and students.

Projects

Departure Brief | *TypeScript, React, Fastify, PostgreSQL* live demo | source

- Built and launched a free preflight briefing app combining **5** public aviation data sources, using **Cloudflare Workers** and **KV** to serve live weather at no hosting cost, backed by **899** automated tests.
- Combined fixed rules with a **local LLM (Ollama)** to rank airspace notices by importance, requiring a quoted source for every claim; in a test on **31** real notices the system flagged every critical one at **$0** running cost.
- Caught that the project's AI quality test could not fail: rules settled **22 of 28** cases before the model ran, so a model giving one fixed answer would score **96%**; rebuilt it to grade the model only on what the rules leave open.
- Ran a **7**-part audit closing **15** blockers in **95** findings, including a rules engine giving GO in a thunderstorm.

Hours to Go | *TypeScript, React, Fastify, PostgreSQL* live demo | source

- Built a web app showing pilots their progress toward **9** Transport Canada licences and ratings, citing every rule.
- Added **scrypt**-secured accounts on **PostgreSQL**, keeping guest logbooks on sign-up, with optimistic updates.

Technical Skills

Languages: TypeScript, JavaScript, Python, HTML/CSS, SQL (PostgreSQL, MySQL, MSSQL)
Frameworks: Node.js, Express, Feathers.js, Fastify, React, Angular
Developer Tools: AWS (Lambda, EKS, EC2, RDS, S3, Athena, Firehose, DynamoDB, CloudWatch), Kubernetes, Docker, Terraform, GitLab CI/CD, Cloudflare Workers, k6, Redis, Ollama, OpenAPI
Libraries: Vite, Vitest, Leaflet, pdf.js, Tesseract.js, AWS SDK
Certifications: Private Pilot Licence (Aeroplane), Transport Canada

Education

Wilfrid Laurier University Waterloo, ON
Honours Bachelor of Science in Computer Science